

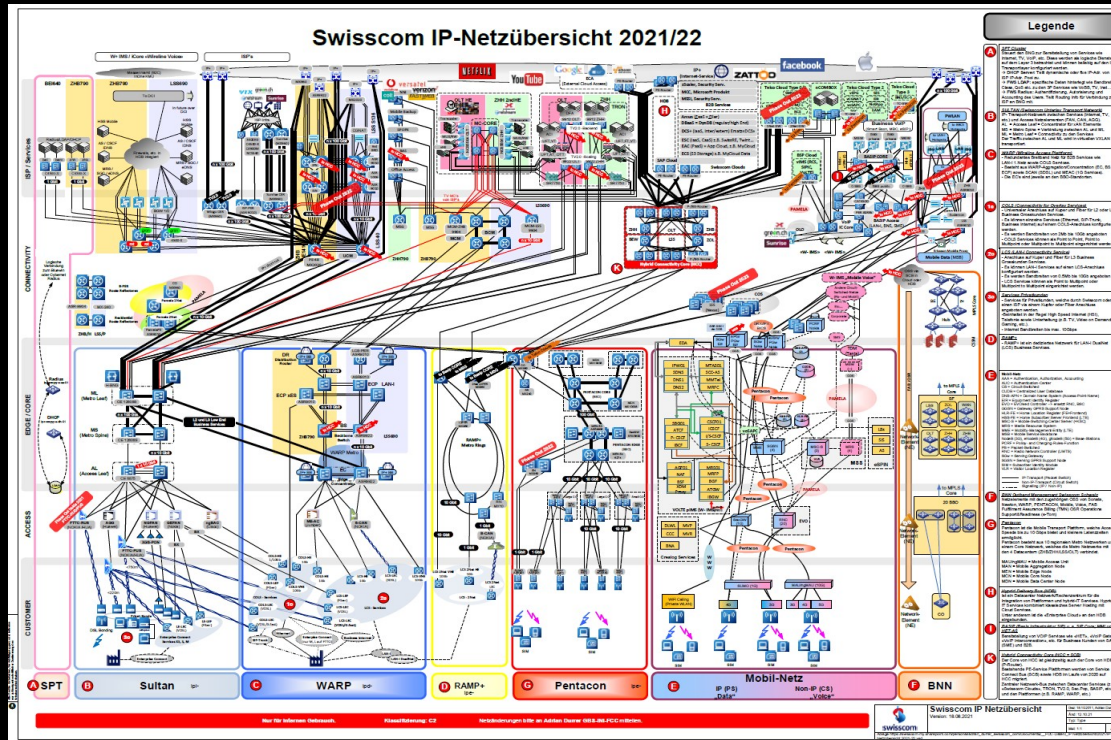
Reaching fully automated backbone networks

a Network Developer's journey

AUTOCON S

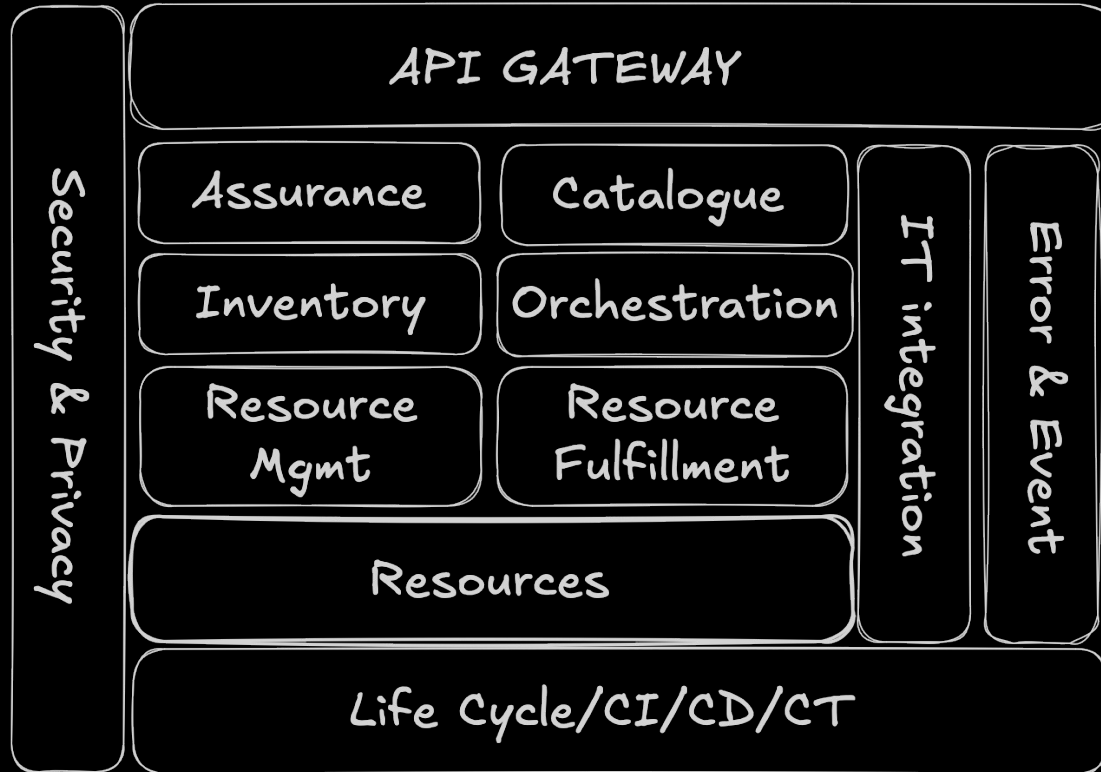
THE NETWORK AUTOMATION CONFERENCE

Holy mess

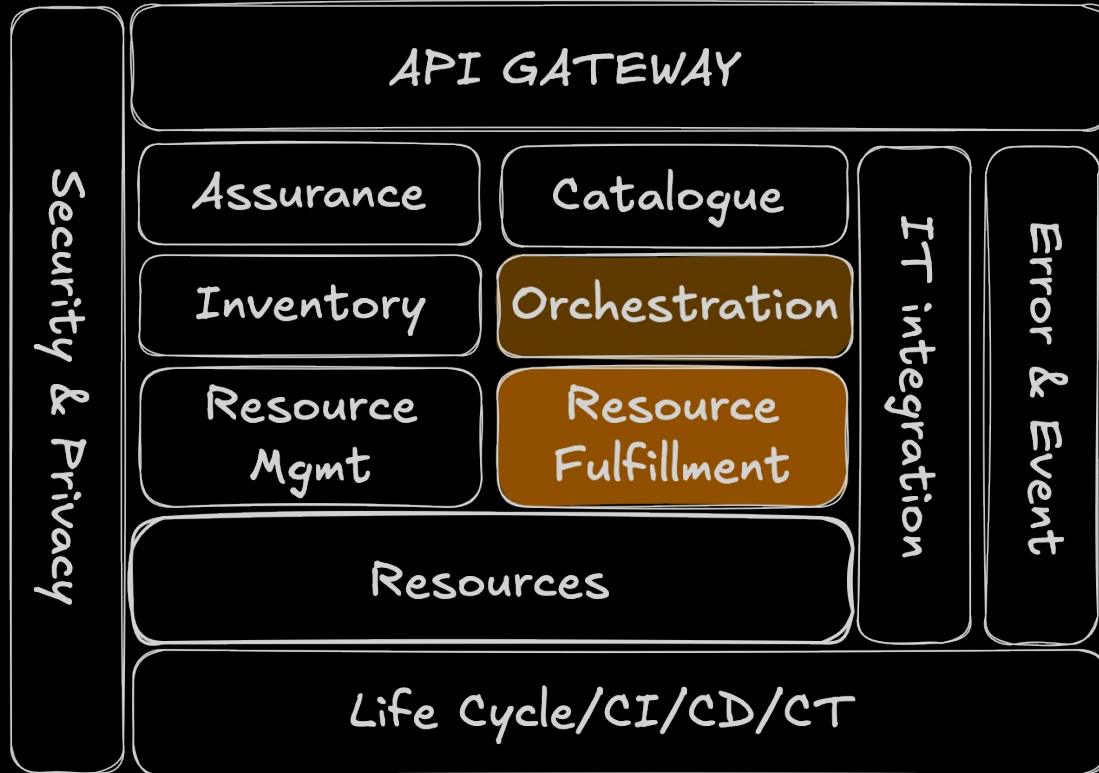


The Vision: Software-Driven Network as a Service

Initial goals

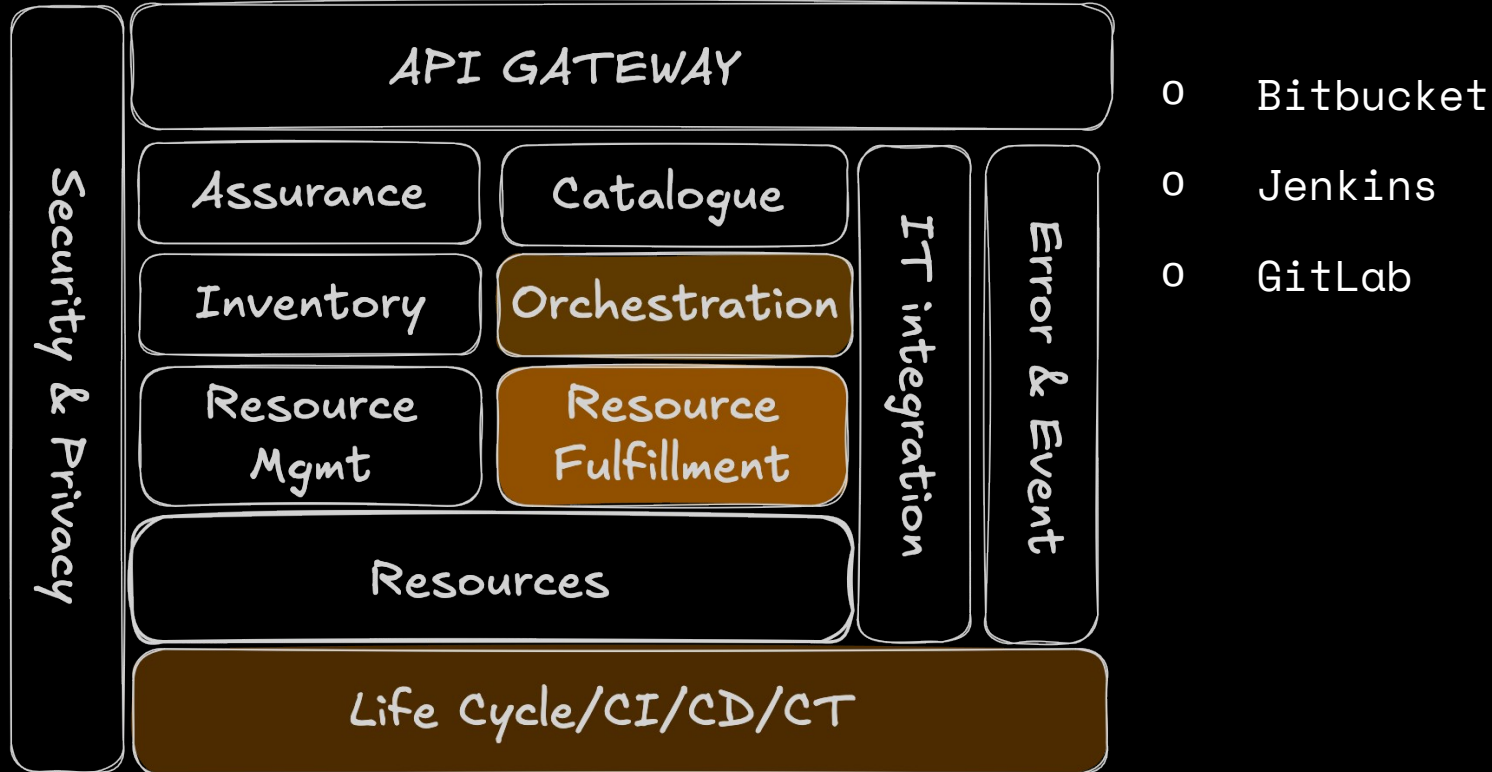


Initial goals

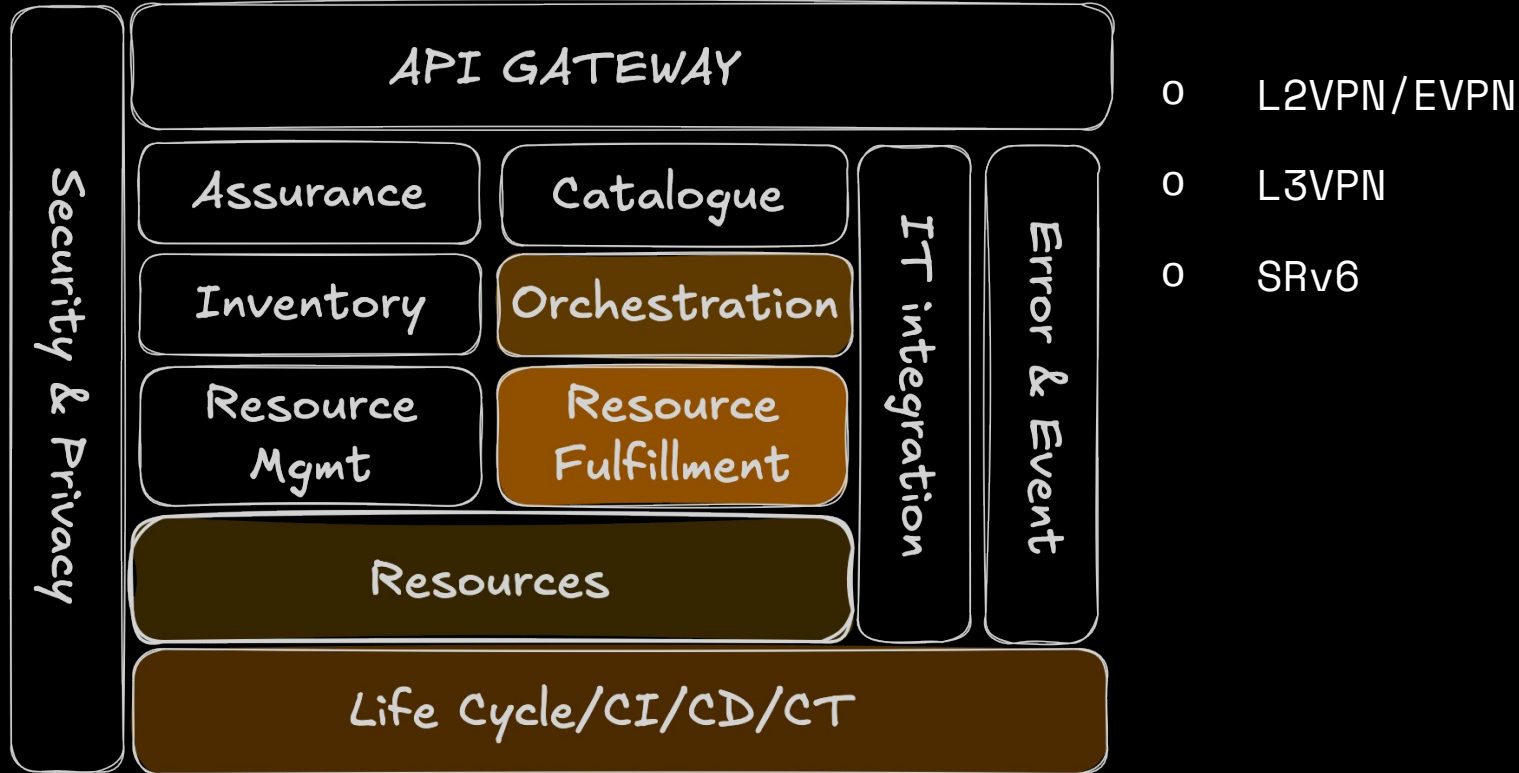


- 0 Thor / CP Broker
- 0 Camunda Workflows
- 0 NSO

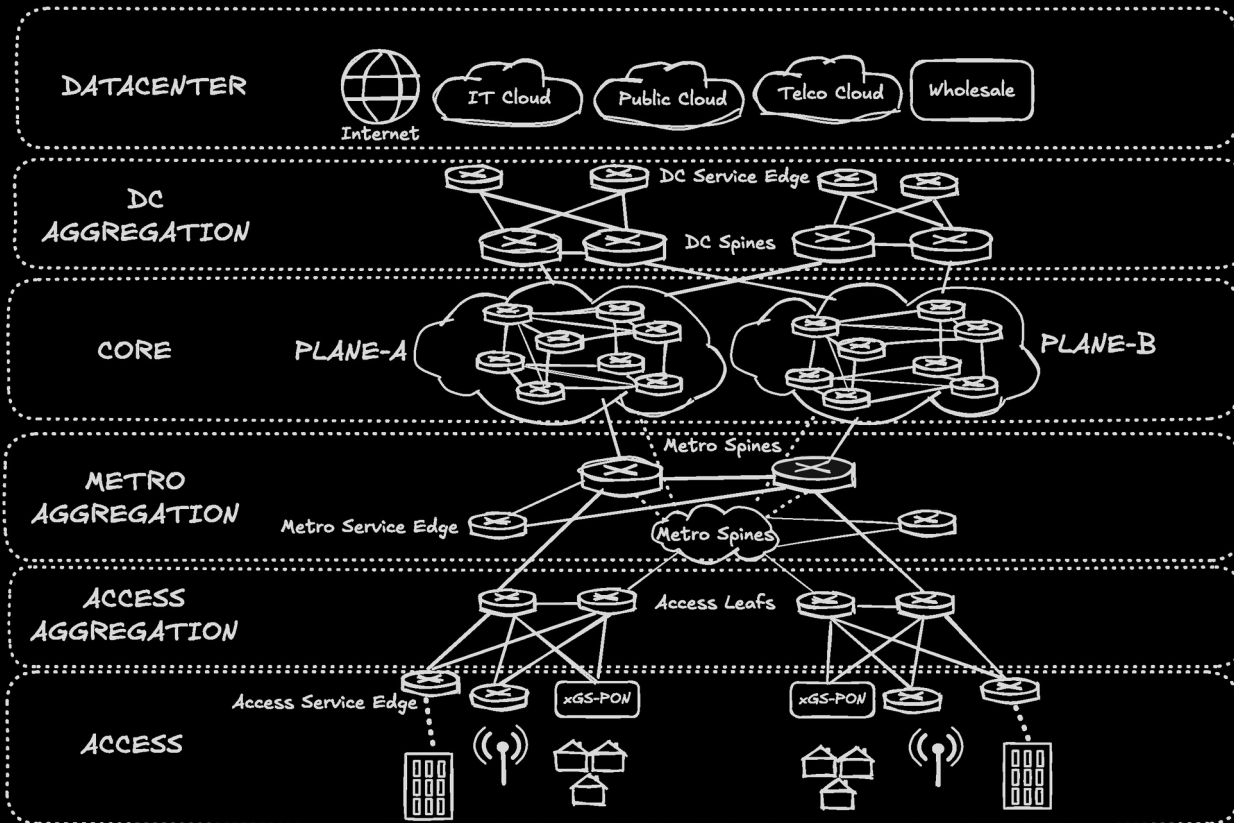
Initial goals



Initial goals

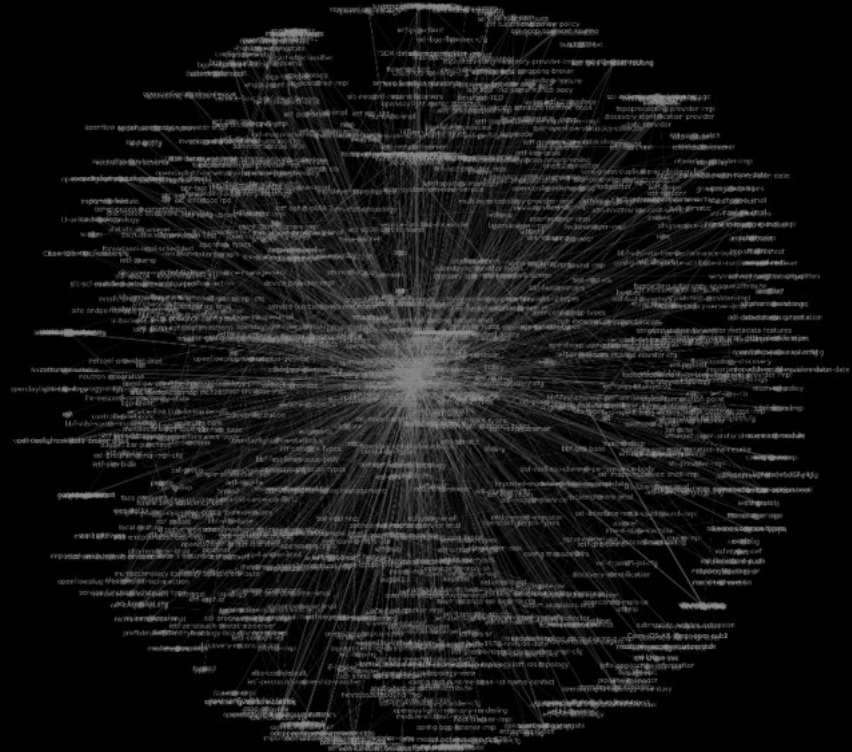


Initial goals



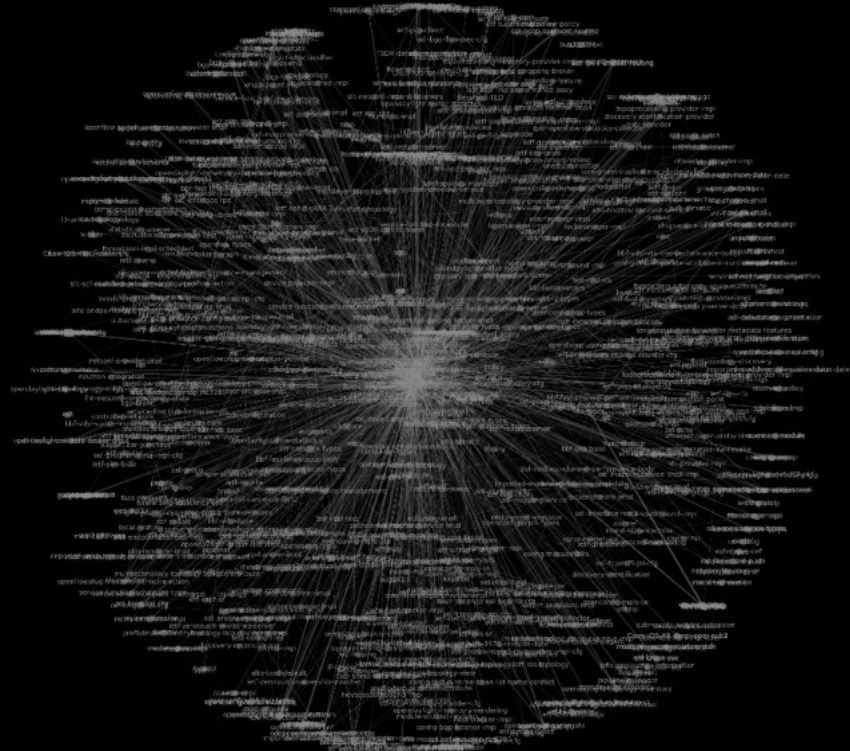
Expected benefits

- o Plug and Play
 - o Each component can be replaced/upgraded independently



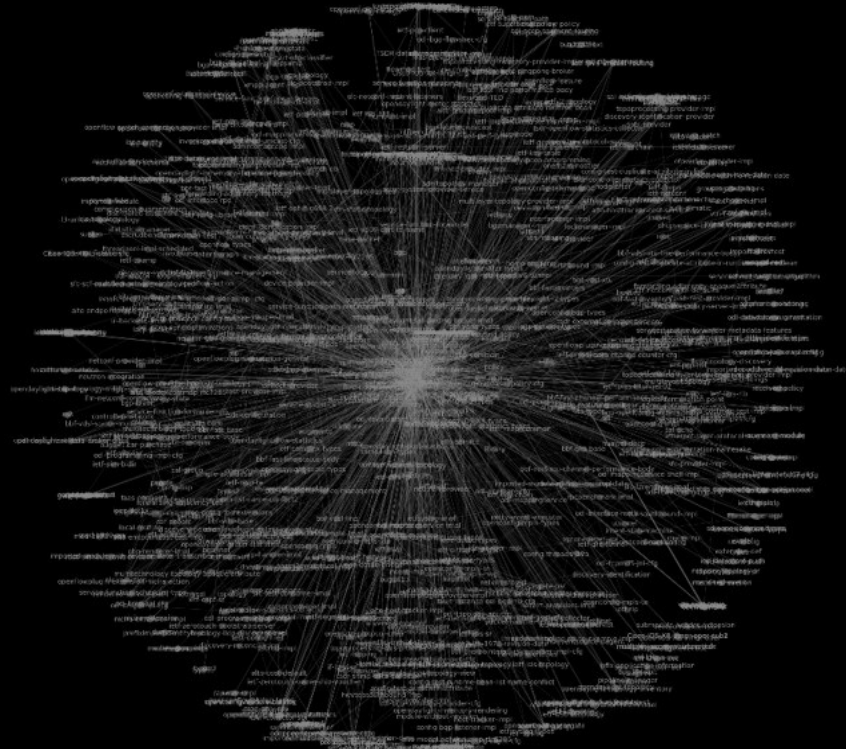
Expected benefits

- o Plug and Play
- o Standardization
 - o Smaller Service Catalog
 - o Standard API



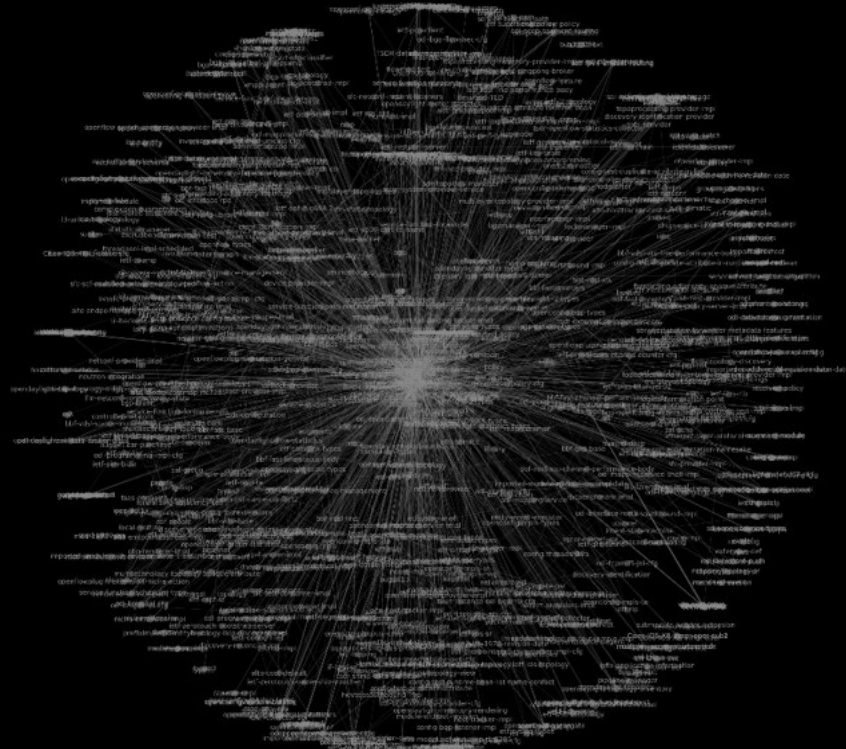
Expected benefits

- o Plug and Play
- o Standardization
- o Time to market
 - o New features and service offers delivered in 2/4 weeks



Expected benefits

- o Plug and Play
- o Standardization
- o Time to market
- o Stability
 - o Less unknowns, faster MTTR



Implementation Realities

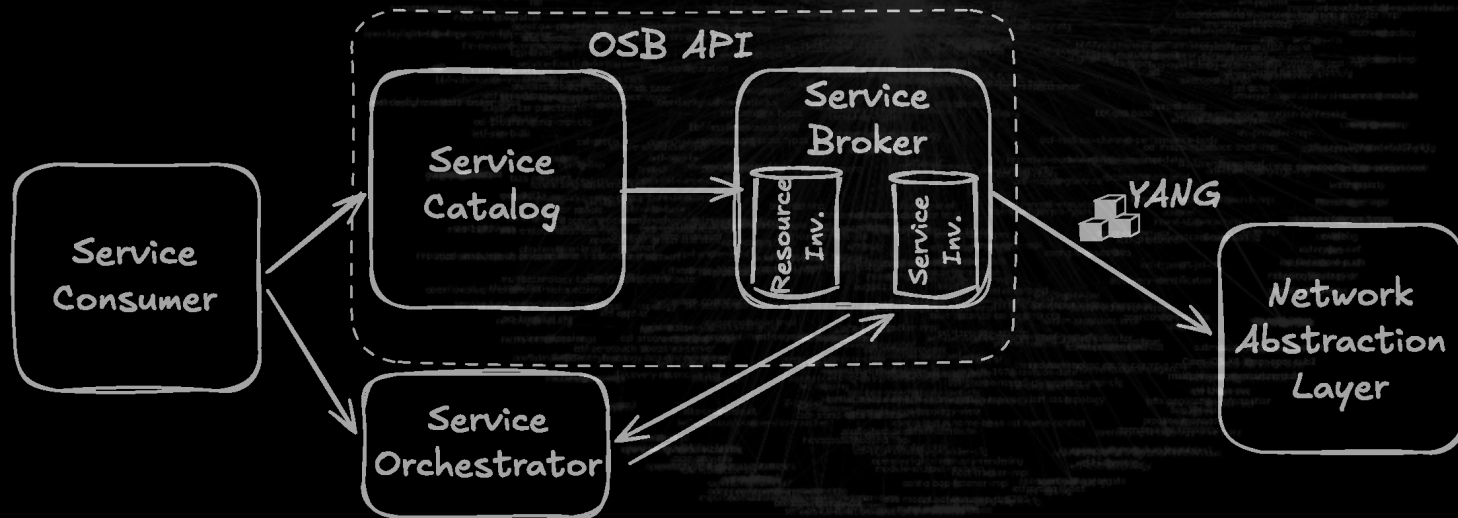
Beginnings and initial hurdles

- o Entering new age



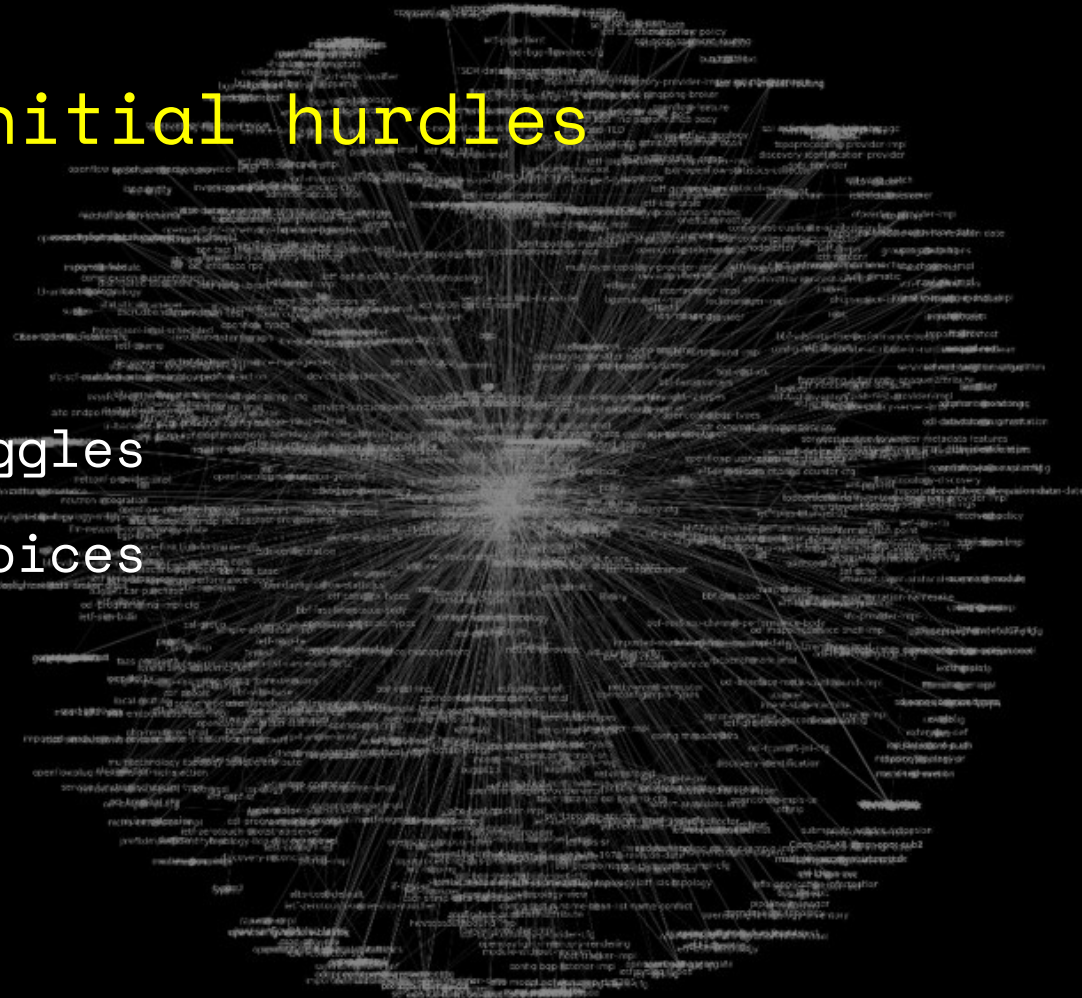
Beginnings and initial hurdles

- o Entering new age
- o Standardization struggles



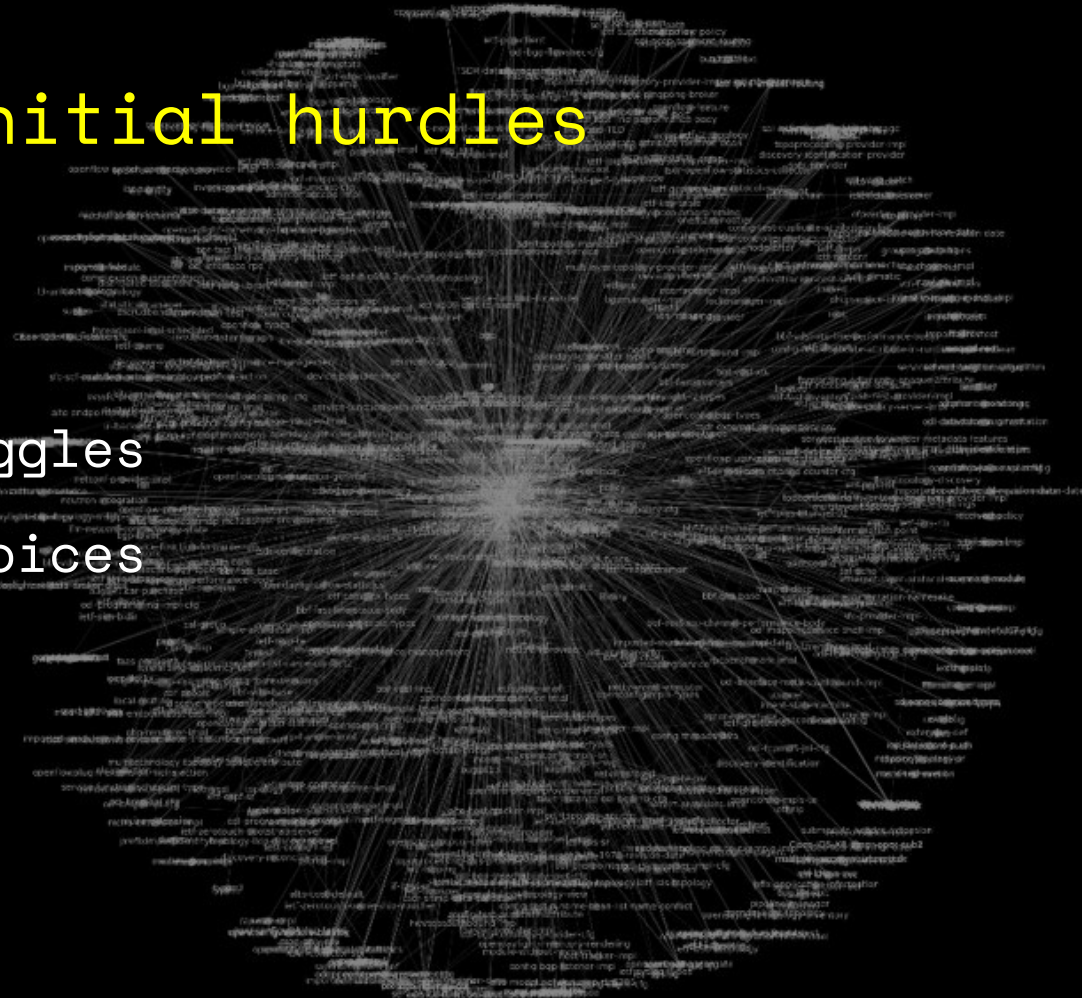
Beginnings and initial hurdles

- o Entering new age
- o Standardization struggles
- o Early engineering choices



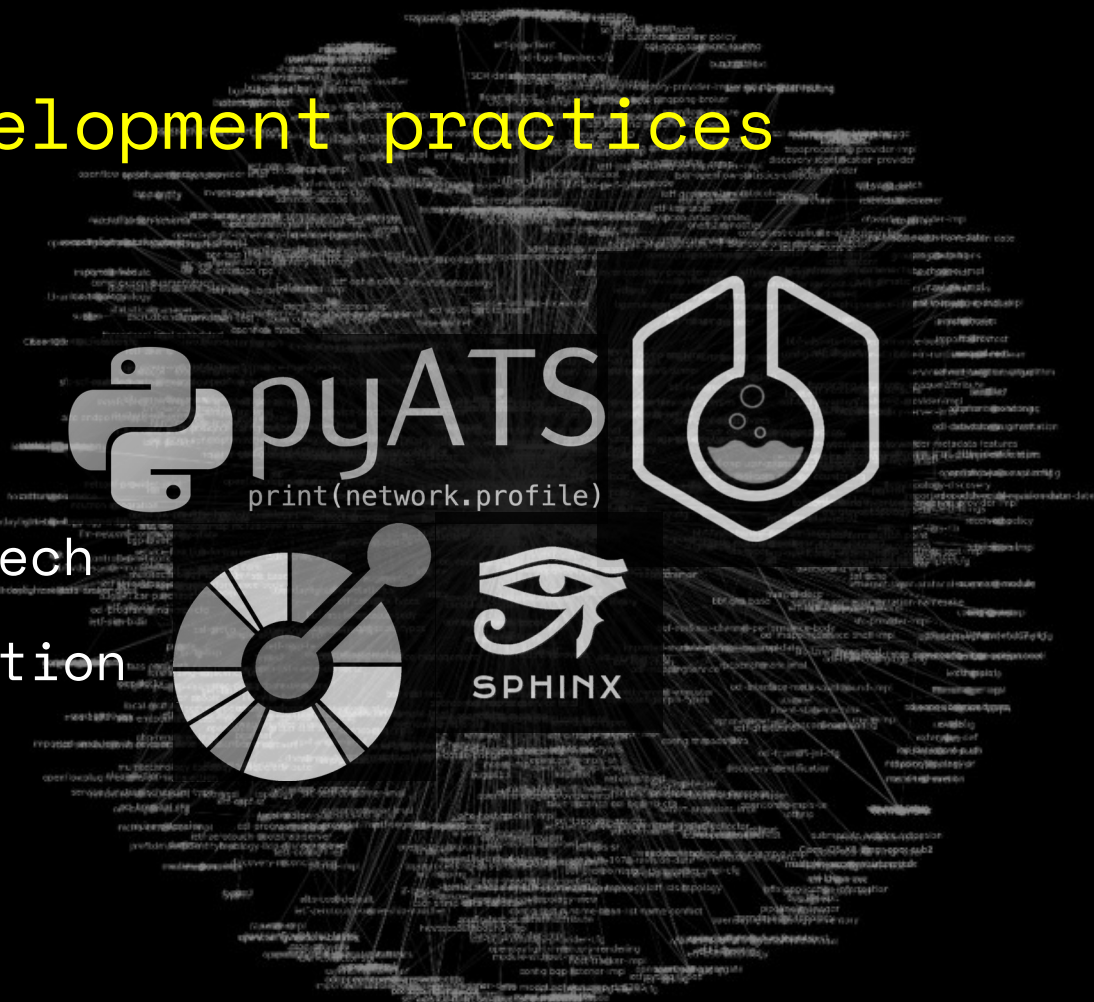
Beginnings and initial hurdles

- o Entering new age
- o Standardization struggles
- o Early engineering choices
- o Toolkit woes



Transforming development practices

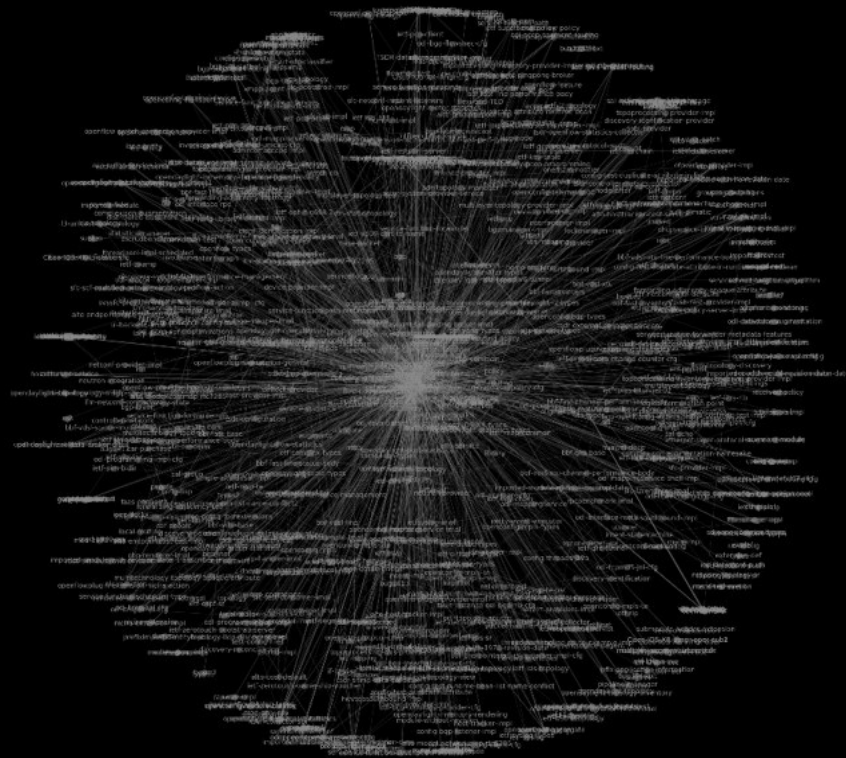
- o Virtual dev stage
 - o repeatable testing
- o Superseding tools/tech
- o Automatic documentation
- o Refined processes



Beyond the Network Stack

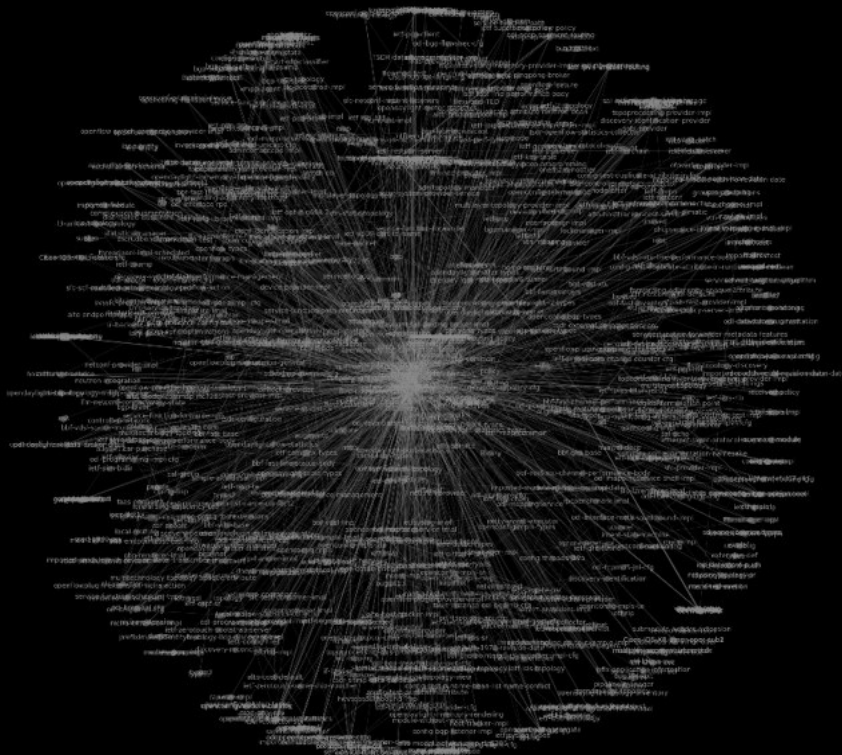
Beyond the Network Stack

- o It's not the network
- o Prototyping vs Test automation



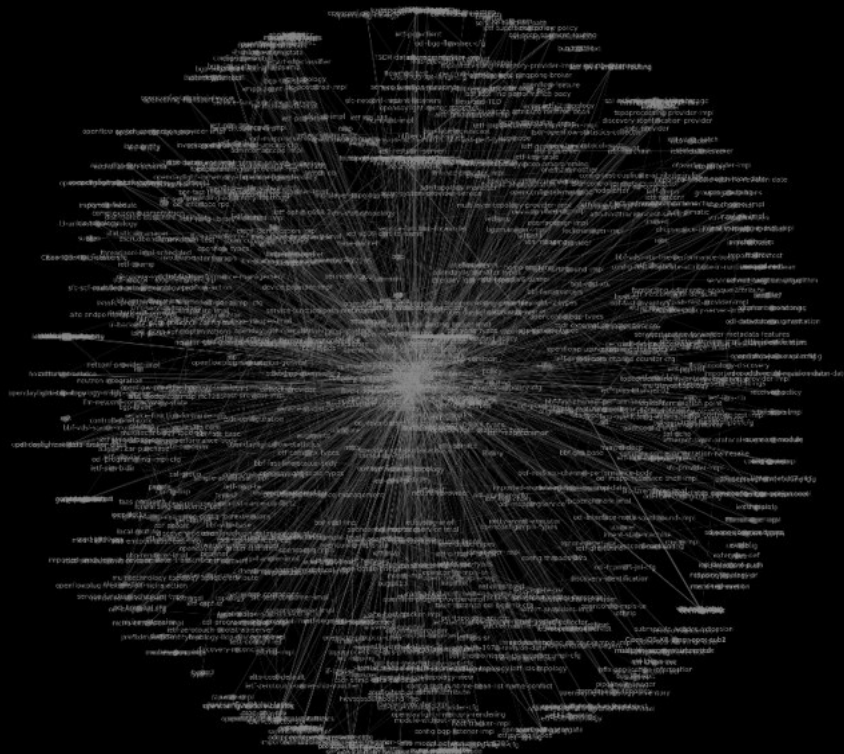
Beyond the Network Stack

- o It's not the network
- o Organizational and cultural hurdles
 - o Dev and Ops
 - o Network Engineer and Developer



Beyond the Network Stack

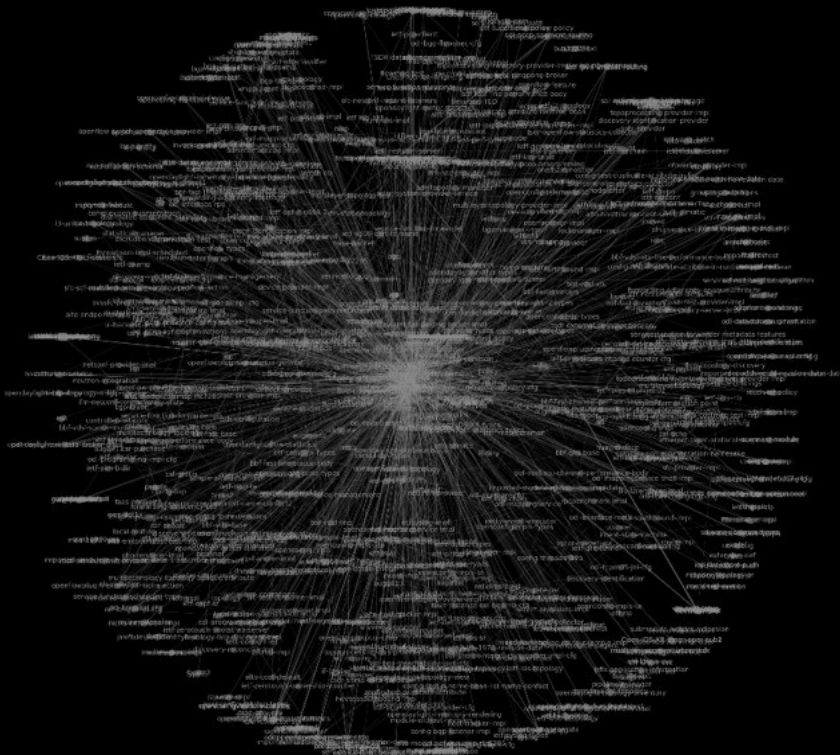
- It's not the network
- Organizational and cultural hurdles
- Gaps between teams and departments
 - Northbound Processes not adapting



Looking Ahead

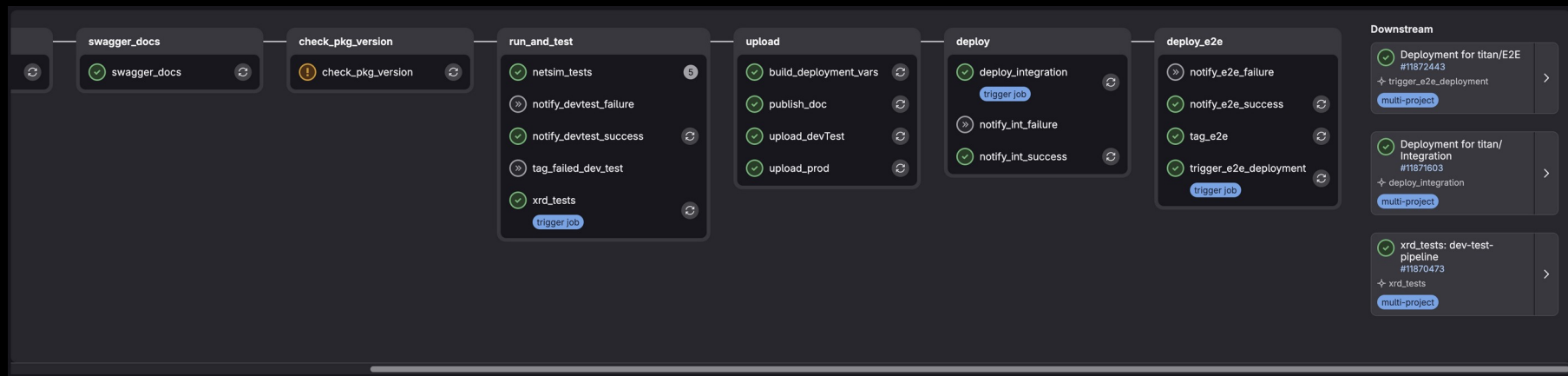
Progress so far

- 0 > 3000 devices (~10k target)
- 0 > 25k instances of public facing services (>150k target)
- 0 0 cli
- 0 0 escalated major incidents



Challenges and lessons learned

0 True CI/CD



Challenges and lessons learned

- 0 True CI/CD
- 0 Scale Testing
 - 0 For loops
 - 0 YANG leafrefs, or other YANG checks
 - 0 Exponential increase in runtime

Snapshot Performance Report

2026-05-29T20:36:10Z — NCS 6.4.0 — Total: 3h 16m

Configuration

batch_size	25
workers	4
timeout	600
shuffle	false
retry_max	3
retry_delay	5
cores	16

titan-cfs-day1

instances	4297
no-diff	78
has-diff	4201
errors	18
stilles	178
retries	0
duration	1h 53m
batch avg	157.52s
batch p95	242.52s
batch max	357.03s

- remote-bundle (14)
- other (4)
- diffs (4201) — 25 patterns

titan-cfs-access-port

instances	4586
no-diff	2041
has-diff	2544
errors	1
retries	0
duration	16m 10s
batch avg	21.06s
batch p95	27.55s
batch max	52.78s

- inventory-missing (1)
- diffs (2544) — 7 patterns

connection-point

instances	29282
no-diff	16700
has-diff	12588
errors	14
retries	87
duration	1h 7m
batch avg	13.82s
batch p95	22.17s
batch max	37.09s

- inventory-missing (14)
- diffs (12588) — 5 patterns

Timeline

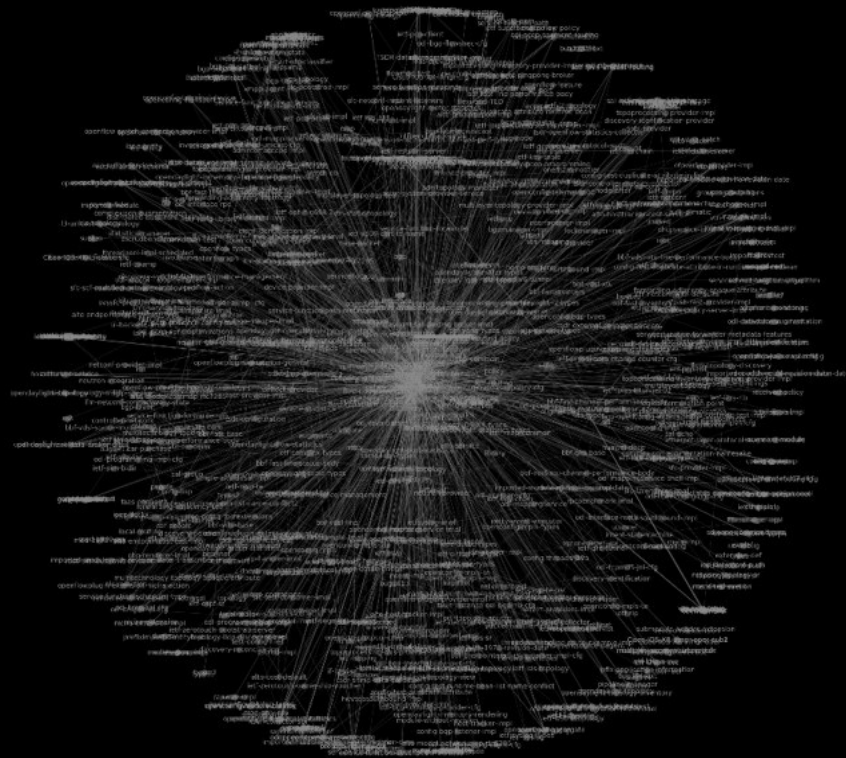


Throughput (instances/min)



Challenges and lessons learned

- o True CI/CD
- o Scale Testing
- o Closed loop/Self healing
 - o Awareness and Analysis
 - o Decision and Execution





Questions?